



North South University
Department of Electrical & Computer Engineering

PROJECT PROPOSAL

Course Code	CSE311L	
Course Title	Database Systems Lab	
Faculty Name	Ms. Tanzilah Noor Shabnam [TnS1]	
Lab Instructor	Md. Amanullah Shah	
Section	5	
Group Number	4	
Date of Submission	13/03/2026	
Group Members Name & ID	Arka Karmoker	2112343042
	Amrita Biswas	2022015642

Project Proposal — Submission Form

1. Project Title & One-Line Description

Title: NSU Transportation Management System

Description: A database-driven web application that will allow NSU students to book daily shuttle tickets, track their bus's live GPS location, and enable admins to easily manage fleets, routes, and schedules.

2. Problem Statement

Currently, NSU relies on a third-party partner (Jatri) to manage its student shuttle service, which limits the university's direct control over the system. Students have to book tickets in advance on a strict first-come, first-serve basis, but they lack a way to track the exact real-time location of their bus, which often leads to missed rides and confusion at bus stops. A dedicated, relational database system is needed to connect students, drivers, and university admins seamlessly, allowing for real-time GPS coordinate updates on map, better seat management (including female-only buses), and full in-house control over ticketing and routing.

3. Core Features / Scope

- **Role-Based Dashboards:** Secure and separate portal interfaces for Admins, Students, and Drivers.
- **Live Bus Tracking:** Drivers can start a ride which pushes their live GPS location to the database, allowing students to track the bus's movement on a real-time map.
- **Dynamic Ticketing System:** Students can search for available routes, select specific times, view seat availability, and book tickets (including a specific option for female-only buses).
- **Fleet & Route Management:** Admins can easily add, update, or delete bus information, set up new destinations with specific map coordinates, and assign drivers to schedules.
- **Ride & Payment History:** Students can track their past and upcoming tickets, while admins can view overall transportation statistics and revenue.

Proposed Tech Stack:

Frontend: HTML, CSS, Bootstrap, JavaScript, jQuery, AJAX

Backend: PHP

Database: MySQL

Libraries: jQuery, DataTables, Leaflet.js, FontAwesome, Bootstrap

APIs: OpenStreetMap, HTML5 Geolocation

Development Tools: XAMPP, VS Code

4. Team Roles

Student Name	Student ID	Role
Arka Karmoker	2112343042	Database Design, Admin Module (Frontend + Backend): 1. Manage Users 2. Manage Buses 3. Manage Routes 4. Manage Rides 5. Manage Payments 6. Track Buses on Map Driver Module (Frontend + Backend): 1. Live Location Sharing, 2. Manage Own Ride 3. Profile Update
Amrita Biswas	2022015642	Student Module (Frontend + Backend): 1. Buy Ticket 2. Ticket History and Status 3. Track Buses on Map 4. View Routes on Map 5. Profile Update Global Module (Frontend + Backend): 1. Signup 2. Login 3. Logout 4. Landing Page (optional) Testing